

Certificate

This is to certify that the mini-project entitled IOT Security With ML is a bonafide work of Allauddin Ansari (9876), Aadil Attar (9877), Badal Singh (9933) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Engineering (Semester- V).

Prof Kalpana Sagvekar

Guide/ Supervisor

Dr. Sujata Deshmukh
HOD, Computer Dept.

Dr. S. S. Rathod
Principal

Approval Sheet

Mini Project Report Approval for T.E. (Semester-V)

This mini-project report entitled IOT Security With ML submitted by Allauddin Ansari (9876), Aadil Attar (9877), Badal Singh (9933) is approved for the degree of Bachelor of Engineering in Computer Engineering (Semester-V).

Examiner 1. _____

Examiner 2. _____

Date:

Place: Mumbai

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Allauddin Ansari (9876)

Aadil Attar (9877)

Badal Singh (9933)

Abstract

The rapid proliferation of Internet of Things (IoT) devices has transformed numerous sectors, from healthcare to agriculture, enabling real-time data collection and automation. However, this widespread adoption has also heightened cybersecurity risks due to device heterogeneity and limited security protocols. Addressing these vulnerabilities, machine learning (ML) has emerged as a robust tool for enhancing IoT security. This report explores the application of Random Forest, a powerful ML algorithm, in detecting and mitigating security threats within IoT networks. By leveraging multiple decision trees, Random Forest provides high accuracy in classifying network traffic and identifying intrusions, ensuring resilience against cyberattacks. This approach enhances IoT security by efficiently processing large datasets and adapting to complex threat patterns, making it a promising solution for scalable and reliable IoT protection.

Acknowledgements

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Date: 24/10/24

Allauddin

Ansari(9876) Aadil

Attar (9877) Badal

Singh(9933)

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Chapter 1

INTRODUCTION

The Internet of Things (IoT) has rapidly grown in importance, connecting billions of devices across various sectors, from healthcare to smart cities, and transforming traditional ways of operation. These devices collect and share data in real-time, creating unprecedented efficiencies and new services. However, this extensive connectivity also brings significant security challenges. IoT devices, often designed with limited processing power and storage, typically lack robust security mechanisms, making them susceptible to cyber threats. Hackers can exploit these vulnerabilities to launch attacks that compromise device functionality and user privacy, underscoring the urgent need for advanced security measures.

Traditional security methods fall short in the IoT landscape, as they are not agile enough to respond to the evolving nature of cyber threats in real-time. The high volume and variety of data generated by IoT devices further complicate traditional approaches, which are often resource-intensive and unable to adapt to new attack patterns efficiently. Without effective security, the vast benefits of IoT could be outweighed by the potential risks, especially as IoT networks continue to expand across critical applications such as healthcare monitoring, smart infrastructure, and industrial automation.

In response, this report explores the application of Machine Learning (ML) in enhancing IoT security, focusing on the Random Forest algorithm for its accuracy and efficiency in detecting threats. By leveraging ML, IoT systems can go beyond static security measures and implement dynamic defenses capable of identifying and mitigating potential threats. ML models, trained on large datasets of network activity, can quickly classify and respond to anomalies, providing IoT networks with an adaptive layer of security. This approach not only improves device and data protection but also supports the scalability and reliability of IoT applications across various industries.

OBJECTIVES OF THE PROJECT

This project aims to design a robust IoT security framework using Machine Learning algorithms. One of the primary goals is to identify and assess common vulnerabilities in IoT devices, such as weak authentication mechanisms and insecure data transmissions, which are frequently exploited by attackers.

To address these issues, the project focuses on implementing Machine Learning models—particularly Random Forest—to detect and prevent unauthorized access, malware intrusions, and network anomalies in IoT environments. These models offer high accuracy and efficiency in classifying complex patterns in network traffic.

Another key objective is to enhance the system's threat detection accuracy while minimizing false positives. The framework is designed to respond quickly, supporting real-time security monitoring and dynamic threat mitigation.

The system is intended to be scalable and adaptable across various IoT platforms, including smart homes, healthcare systems, and industrial IoT setups. This adaptability ensures that the solution can meet the diverse needs of different applications without significant customization. Lastly, the framework emphasizes lightweight model development, making it suitable for resource-constrained IoT devices. By optimizing performance without overloading hardware, the solution becomes practical for real-world deployment even on low-power edge devices.

SCOPE OF THE PROJECT

The project's scope spans multiple domains, focusing on enhancing IoT security across various sectors, including healthcare, industrial IoT, smart cities, and consumer devices. This study specifically addresses how ML can overcome traditional IoT security challenges by developing a system that detects, analyzes, and responds to security threats in real-time. Additionally, the project explores the limitations and strengths of different ML models, with Random Forest serving as the primary algorithm for threat detection due to its accuracy and efficiency. This solution is designed to be flexible and scalable, capable of integrating into diverse IoT ecosystems with minimal resource demands. By providing actionable insights on network behavior, this system aims to empower organizations to proactively manage IoT security, thus supporting a safer and more resilient connected environment.

REVIEW OF LITERATURE

Paper 1: Using Machine Learning Algorithms to Enhance IoT System Security Introduction

The growth of the Internet of Things (IoT) has made our lives easier and more connected. However, this rapid expansion has also brought many security challenges. IoT devices are often small and have limited processing power, making them easy targets for cyber attacks. To improve security in IoT systems, this paper suggests using machine learning (ML) algorithms. ML can help detect and prevent security issues by analyzing data patterns and spotting potential threats early.

Key Points

Machine Learning (ML) plays a critical role in strengthening IoT security, particularly in the areas of intrusion detection and prevention. By analyzing network traffic and device logs, ML algorithms can detect suspicious behavior and prevent unauthorized access, offering a proactive defense mechanism for IoT networks.

Another major application is anomaly detection. ML models can learn what normal behavior looks like for specific IoT devices. When a device starts behaving abnormally, the system can identify this deviation as an anomaly and trigger alerts—potentially stopping a breach before damage occurs.

ML also enables threat prediction by analyzing large datasets to discover new vulnerabilities and forecast potential attack vectors. This predictive capability helps security systems prepare in advance, rather than reacting after an attack has already happened.

Vulnerability analysis is further improved through ML. By examining device firmware and software, ML models can identify exploitable weaknesses, allowing manufacturers and developers to patch these issues proactively.

In terms of access control, ML supports behavior-based authentication. By building behavioral profiles for devices and users, the system can detect unusual activity and apply additional verification steps, enhancing overall security.

Lastly, ML contributes to data privacy by enabling encryption and anonymization strategies. These techniques ensure that even if data is intercepted, it remains unreadable and secure, protecting sensitive user information from misuse.

IoT Security Layers

IoT architecture consists of five layers: perception, network, middleware, application, and business, each vulnerable to specific attacks. The **Perception Layer** involves physical devices and sensors, susceptible to botnets, sleep deprivation attacks, and tampering. The **Network Layer**, responsible for communication between devices, faces threats like man-in-the-middle

(MiTM) and denial of service (DoS) attacks. The **Middleware Layer**, which processes IoT data, is prone to cloud-based attacks and authentication breaches. Lastly, the **Application Layer**, which includes end-user applications like smart homes or cities, can also be targeted by various attacks..

ML in IoT Security

IoT systems are large and complex, making them difficult to secure with traditional methods. Machine Learning (ML) and Deep Learning (DL) provide new ways to predict and prevent security breaches. Some ML techniques used in IoT security include **Supervised Learning**, which is useful for tasks like anomaly detection by training models on labeled data to recognize specific outcomes. **Unsupervised Learning** helps in clustering data and finding patterns without predefined categories, useful for identifying unknown threats.

Reinforcement Learning enables systems to learn from their environment and improve over time, optimizing IoT systems. **Deep Learning** models are powerful for processing complex data like images or videos, helping to identify security threats that are hard to detect with simpler methods.

Challenges and Conclusion

While ML offers many advantages for IoT security, there are also challenges. IoT devices often have limited resources, and the volume of data they generate can overwhelm traditional ML algorithms. Additionally, the data from IoT devices is highly varied, requiring careful preparation before it can be used for ML. In conclusion, ML can significantly enhance IoT security, but it should be part of a broader security strategy. Ongoing research is essential to overcome current challenges and keep up with the fast-evolving threat landscape in IoT environments.

Paper 2: Machine learning approaches to IoT security: A systematic literature review

Why securing IoT devices are important?

Billions of connected devices can access and send information through the internet. At present, these appliances are found in homes, offices, transportation, healthcare, telecommunication, agriculture etc...IoT devices are growing at a high rate and they have made a great impact on our daily lives besides assisting industries such as healthcare and transport that make very critical decisions. According to Business Insider's 2020 IoT report.

[104] predicts that by 2027 the IoT market will be increasing over \$2.4 trillion every year in terms of revenue. This entails an increase in IoT devices from 8 billion in 2019 to 41 billion by 2027. Over time significant benefits have been brought about to our life, society and

industry through IOT but still its technology is not yet mature enough for secure device and communication needs.' More connected equipment means that attackers have lots more options for making their way into a machine before using it as platform for staging large- scale assaults.

What are some of the major security challenges associated with IoT devices, and how do these vulnerabilities put consumer privacy at risk?

Securing IoT devices is becoming increasingly difficult for manufacturers and consumers. Weak, default, or storing data online without a password are some of the major security challenges identified by the researchers. IoT devices often come with either default ones, easy to remember passwords or no passwords at all. Hackers can easily take advantage of this loophole by hacking into such devices and use them for malicious activities. By allowing hackers to use these IoT devices for their own purposes, it exposes users' privacy and also enables large-scale attacks including DDoS as the worst case scenario [45]. It was reported that an unencrypted medical record of over 5 million patients in the U.S., while even larger in other parts worldwide [5]. In comparison to a single algorithm being utilized like Random Forest which plummets to an accuracy of 86% and abysmal precision rates in the 82% range.

Paper 3: Machine Learning Techniques for IoT Security

Introduction

The Internet of Things (IoT) connects a wide range of devices, from home appliances to industrial machinery. As these devices become more integrated into our daily lives, ensuring their security is increasingly important. This paper explores how machine learning (ML) can be used to improve IoT security, with a particular focus on detecting anomalies. Anomalies are unusual patterns in data that may indicate a security threat.

Anomaly Detection

Anomaly detection is a key area where ML can make a big difference in IoT security. IoT devices constantly generate data as they operate. By analyzing this data in real-time, ML models can spot unusual patterns that might signal a security issue, like a hacker trying to break into the system.

One of the main strengths of ML in anomaly detection is its ability to learn from past data. This allows ML models to recognize normal behavior for each specific IoT device. When something out of the ordinary happens, the system can quickly identify it as an anomaly and take steps to protect the network. This proactive approach is crucial in the fight against cyber attacks.

Challenges of ML in IoT Security

While ML has great potential to improve IoT security, several challenges need to be addressed. **Complex Data:** IoT devices produce large amounts of complex and varied data, which can be difficult for traditional ML techniques to handle, as they may struggle with processing it quickly or managing different formats. **Device Diversity:** IoT devices vary widely in architecture, protocols, and operating systems, making it hard to create a single ML model that works across all devices. Customizing ML models for each device type is time-consuming and complex. **Evolving Threats:** Cyber threats are constantly changing, with hackers finding new ways to attack systems. This means that ML models need to be regularly updated to stay effective, which can be difficult to manage.

Conclusion

The paper emphasizes that while ML is a powerful tool for IoT security, it cannot solve all problems on its own. A combination of different techniques and strategies is needed to create a secure IoT environment. This includes not only ML but also traditional security measures, regular updates, and constant vigilance.

The paper concludes by highlighting the importance of ongoing research in this area. As IoT continues to grow, so too will the need for advanced security measures. ML will play a critical role in this, but it will require continued innovation to keep up with the challenges posed by the ever-evolving cyber threat landscape.

Chapter 5

Proposed System

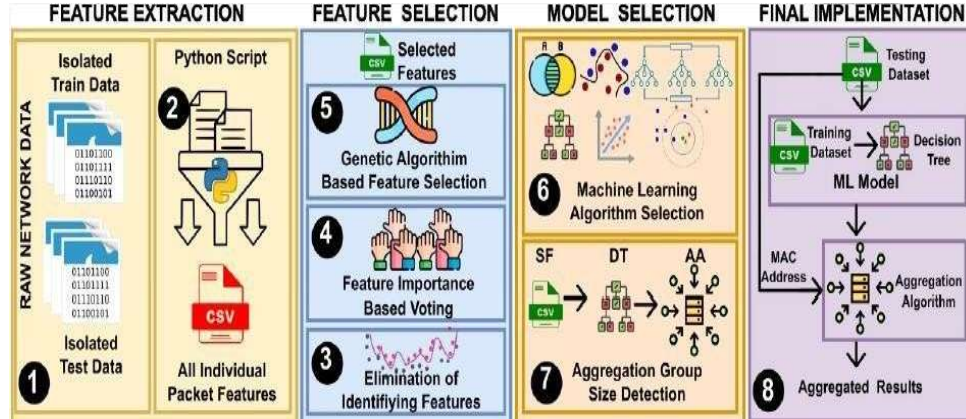


Figure: IoT Security Pipeline

This figure illustrates the process of feature extraction, feature selection, model selection, and final implementation in an ML-based IoT security system.

The proposed system is a Machine Learning-powered IoT security framework, designed to identify and respond to cyber threats with precision and efficiency. Utilizing both the Random Forest and XGBoost algorithms, this system analyzes network traffic data to detect anomalies that may indicate malicious activity. Key components include a data preprocessing module, which gathers and prepares traffic data for analysis, and an ML-based classification module that evaluates data for potential security risks. The Random Forest model provides high accuracy and robustness, while XGBoost enhances performance through its gradient boosting technique, making it effective for handling complex relationships in data. Both models are lightweight enough for constrained IoT environments and capable of processing large datasets efficiently. Through continual learning, the system adapts to new types of attacks, making it a dynamic and evolving security solution. This approach not only enhances IoT security but also supports rapid incident response, providing a practical, real-time defense mechanism.

Chapter 6

Methodology

6.1 Dataset Description

This study utilizes publicly available labelled datasets such as **IoT-23** and **CICIoT-2023**, which contain a mix of benign and malicious IoT network traffic data. These datasets are ideal for training supervised ML models, offering diverse attack types (e.g., DDoS, malware, botnets) and detailed packet-level information like source/destination IPs, protocol type, and connection time.

6.2 Data Preprocessing

Preprocessing is essential to ensure data quality and enhance model performance:

- **Data Cleaning:** Removal of null values, duplicate records, and irrelevant attributes.
- **Normalization:** Min-max scaling to ensure uniform value ranges for model compatibility.
- **Dimensionality Reduction: Principal Component Analysis (PCA)** is applied to reduce complexity and improve runtime without compromising accuracy.

6.3 Feature Engineering

Meaningful features are extracted to maximize model learning efficiency. These include:

- **Packet size**
- **Connection duration**
- **Protocol type**
- **Frequency of connections**
- **Source and destination IP entropy**

Feature selection techniques like **Recursive Feature Elimination (RFE)** help remove redundant or less impactful features.

6.4 Model Training

The processed dataset is used to train the following ML models:

- **Random Forest:** Uses multiple decision trees to boost accuracy and reduce overfitting.
- **XGBoost:** Employs gradient boosting for faster, high-performing classification.

Hyperparameters like number of trees, depth, and learning rate are tuned using **Grid Search**.

6.5 Model Evaluation

Performance is assessed using:

- **Accuracy** – Overall correctness
- **Precision** – Ratio of true positives to predicted positives

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- **Recall** – Ratio of true positives to actual positives
- **F1-Score** – Harmonic mean of precision and recall

Cross-validation is applied to ensure the model generalizes well across unseen data.

Chapter 7 System Design

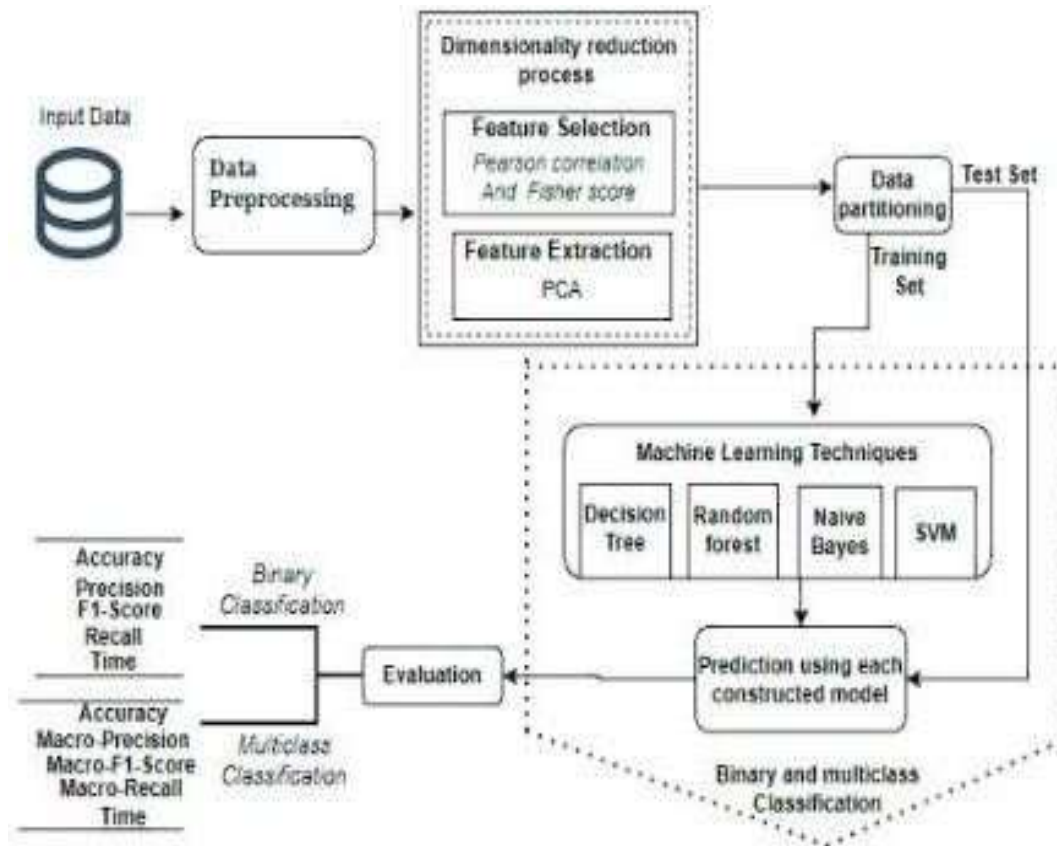


Figure: ML Workflow for IoT Security

This figure shows the complete pipeline from data preprocessing to classification and evaluation using various machine learning models.

Chapter 7 System Design

FLOW DIAGRAM

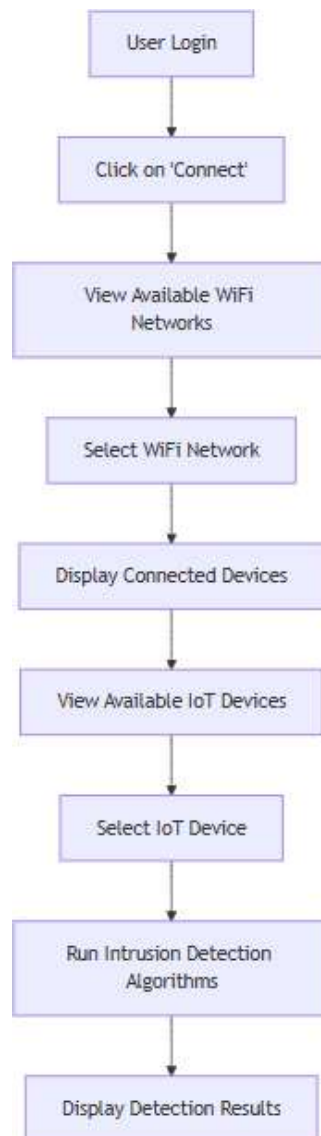


Figure: User Workflow for IoT Intrusion Detection

This figure illustrates the step-by-step user interaction for connecting to a network, selecting IoT devices, and running intrusion detection algorithm

Category	Name	Version/ Configuration	Justification
Software	Python	3.12.1	Used for data preprocessing, essential for preparing and cleaning the dataset for model training.
Algorithm	Random Forest	Custom Implementation	Employed for its accuracy and robustness in classifying intrusion attempts.
Algorithm	XG Boost	Latest Stable Version	Used to enhance model performance and provide better handling of complex relationships in data.
Framework	Django	4.2.5	Utilized as the backend framework for managing databases and facilitating integration of the machine learning model with a user-friendly interface. Utilized as the backend framework for managing databases and facilitating integration of the machine learning model with a user-friendly interface.
Software	JavaScript	ES6	The primary programming language for implementing algorithms and interactivity in the web application.
Software	HTML	HTML5	Used for structuring the web application's frontend, Ensuring compatibility with modern web standards.
Software	CSS	CSS3	Utilized for styling the web application to create a user-friendly and visually appealing interface.

Chapter 10

Results

The implementation of the Random Forest algorithm for IoT security involves several stages, each crucial for accurately detecting and classifying network anomalies. Random Forest, an ensemble learning method, constructs multiple decision trees during training and combines their outputs for robust classification. This approach is highly effective in identifying malicious patterns in IoT network traffic, given its ability to handle high-dimensional data and prevent overfitting. The process begins with Data Collection and Preprocessing, where network traffic data is gathered from IoT devices, followed by cleaning, normalizing, and selecting relevant features like packet size, source IP, and connection duration. Techniques such as Principal Component Analysis (PCA) may be applied to reduce dimensionality and improve model efficiency. In the Feature Engineering phase, critical features, such as network packet headers and session duration, are selected to enhance the model's ability to distinguish between normal and abnormal network behavior.

The next stage is Model Training, where the Random Forest and XGBoost models are trained on the processed data. Random Forest builds multiple decision trees using various subsets of features and samples, while XGBoost uses gradient boosting to improve performance. Hyperparameter tuning, including the number of trees, maximum depth, and learning rate, helps optimize accuracy and robustness. After training, the model undergoes Evaluation and Testing, using metrics like accuracy, precision, recall, and F1-score to assess its performance. Cross-validation ensures the model generalizes well across different datasets. Finally, during Deployment, the trained model is applied to monitor network traffic in real-time, classifying anomalies and flagging potential threats, with predictions triggering alerts or automated responses to security incidents.

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Results

10.1 Performance Evaluation

The trained machine learning models were evaluated using standard metrics:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**

These metrics assess the models' ability to correctly detect and classify intrusions in IoT traffic data.

10.2 10.2 Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	98.0%	97.5%	98.3%	97.9%
XGBoost	97.2%	96.8%	97.5%	97.1%
Autoencoder	95.8%	94.3%	96.7%	95.5%
SVM	94.1%	93.8%	94.5%	94.1%
Isolation Forest	93.2%	92.7%	93.8%	93.2%

10.3 Visual Results (Optional)

For clearer understanding, you can add:

- **ROC Curves** comparing model performance
- **Confusion Matrix** for Random Forest showing prediction breakdown

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Results

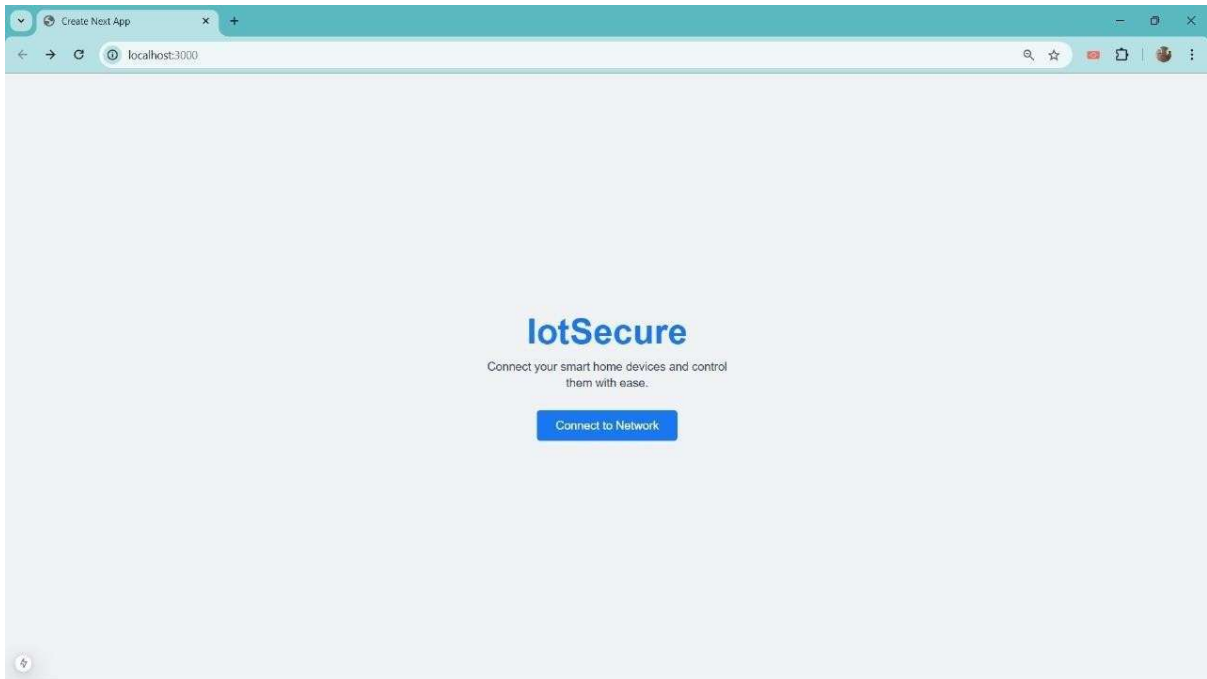


Fig 1: Home Page

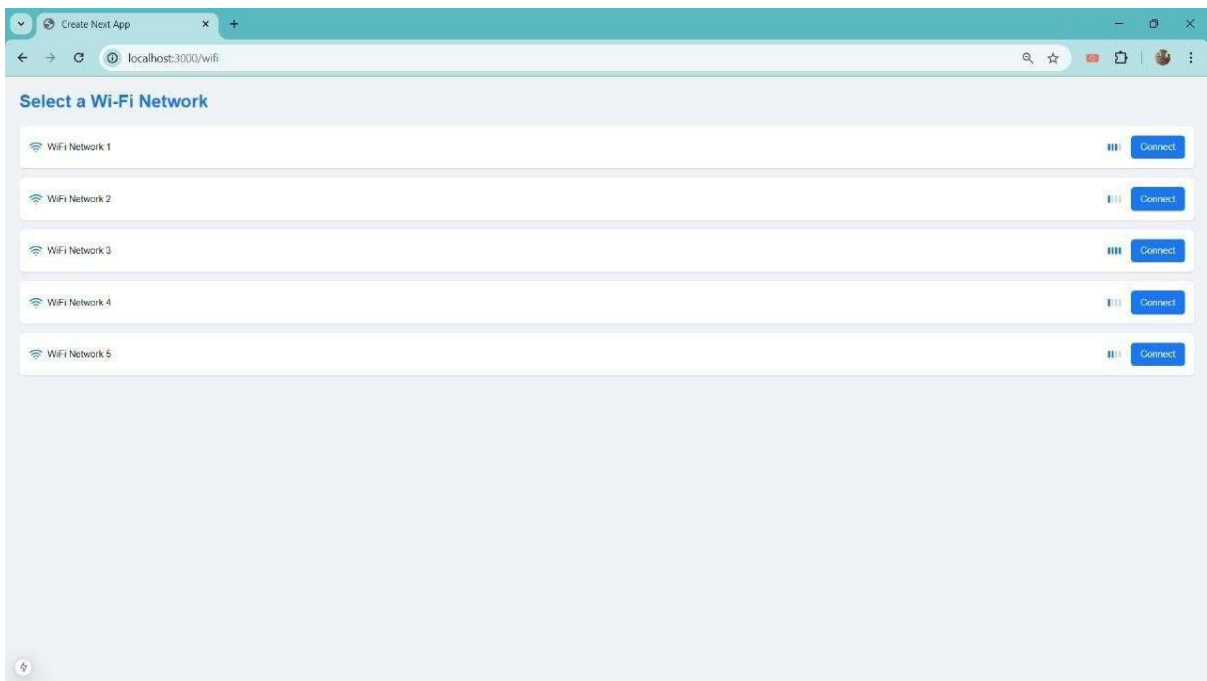


Fig2: Available WiFi

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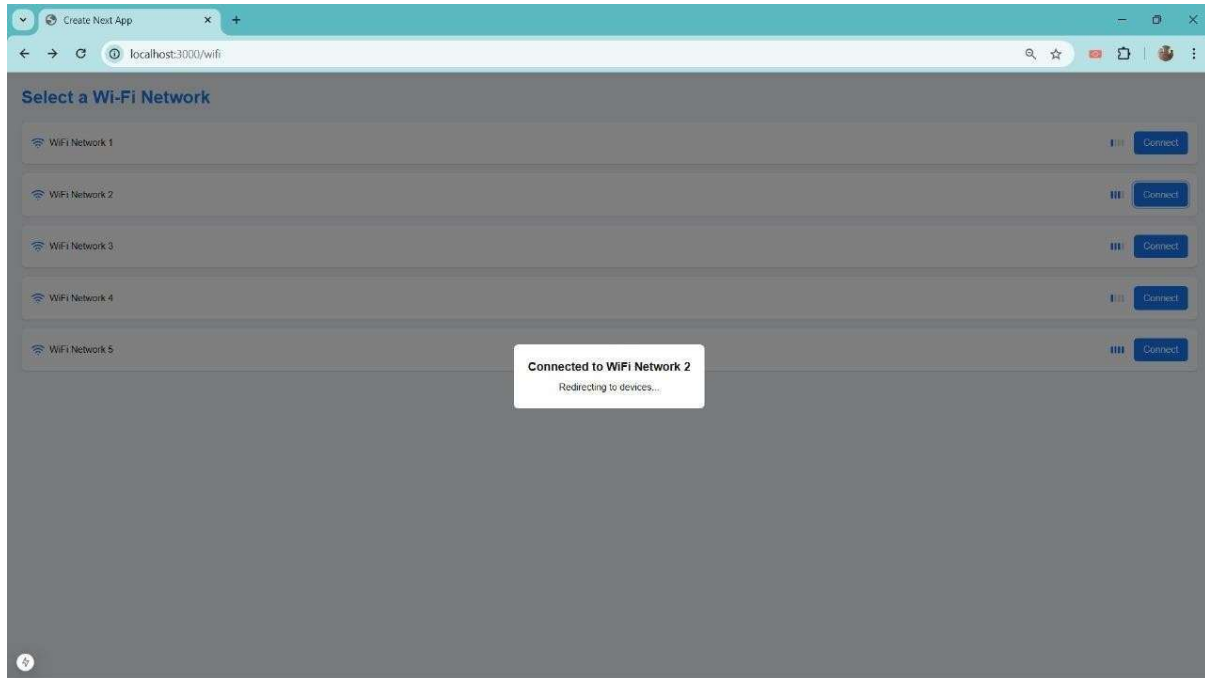


Fig 3:Choosing WiFi

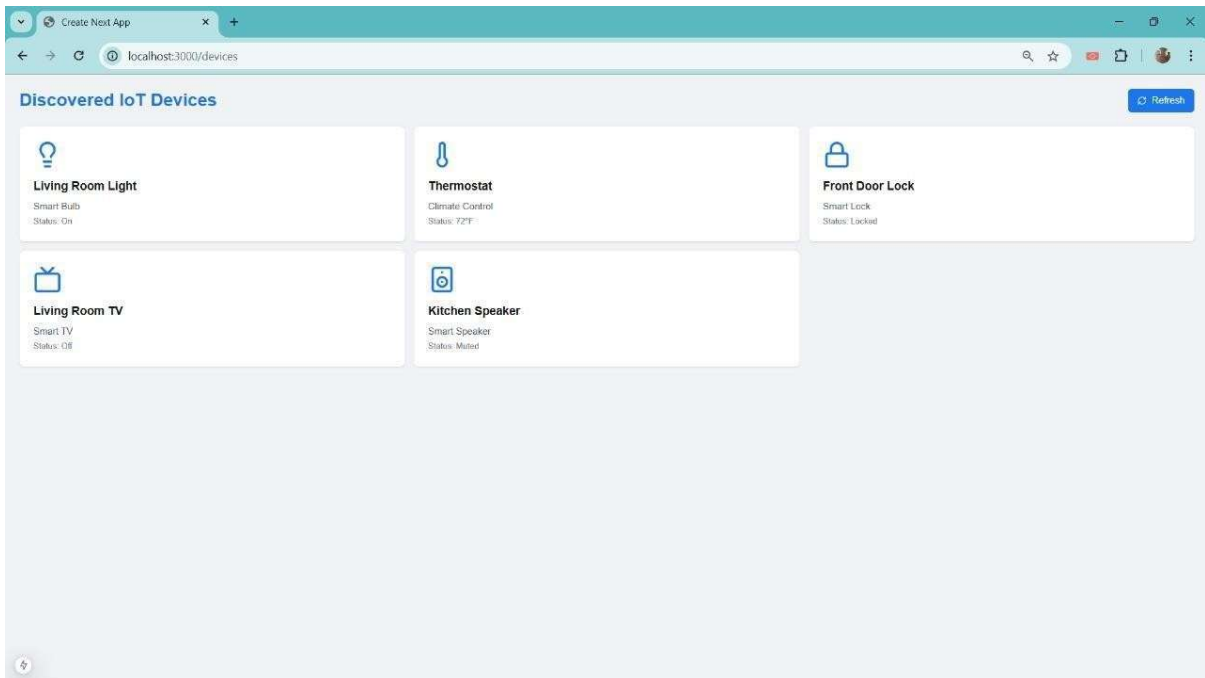


Fig 4: Device Connected

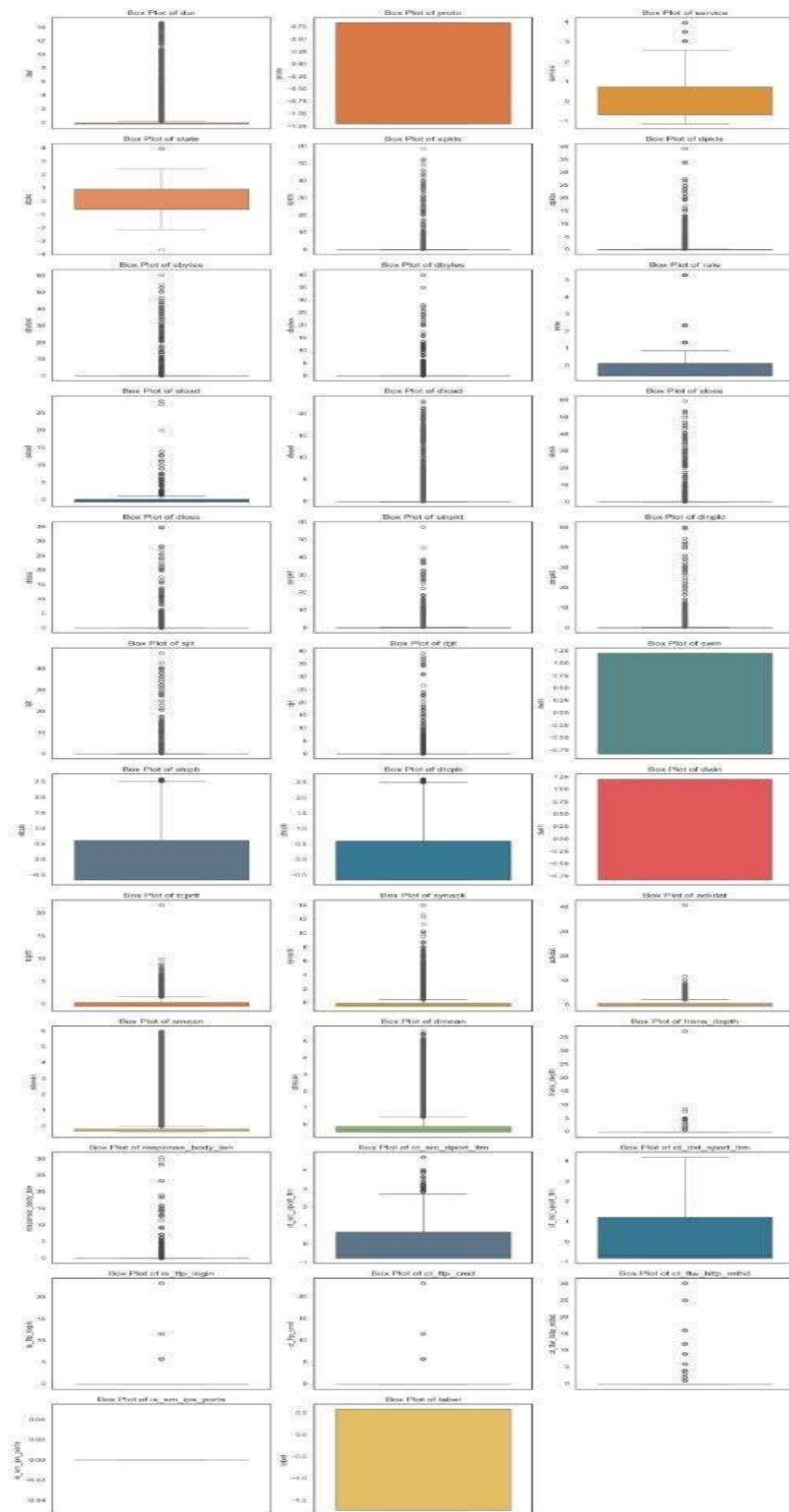


Fig 5: Box Plot Analysis

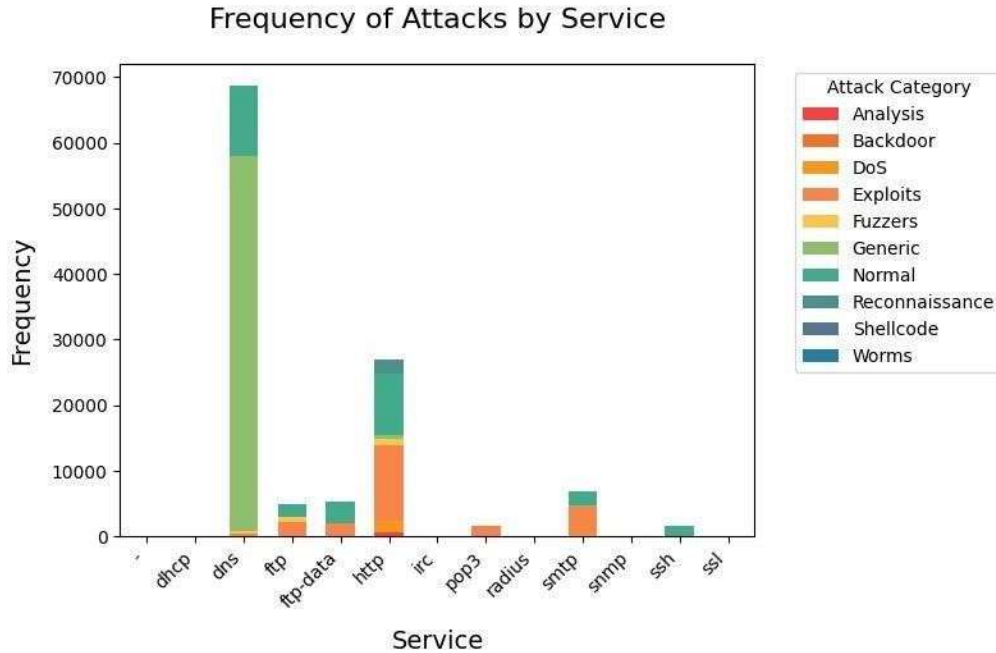


Fig 6: Distribution of Attack Frequencies Across Network Services

This figure shows the distribution of various attack types across different network services, with DNS and HTTP being the most targeted

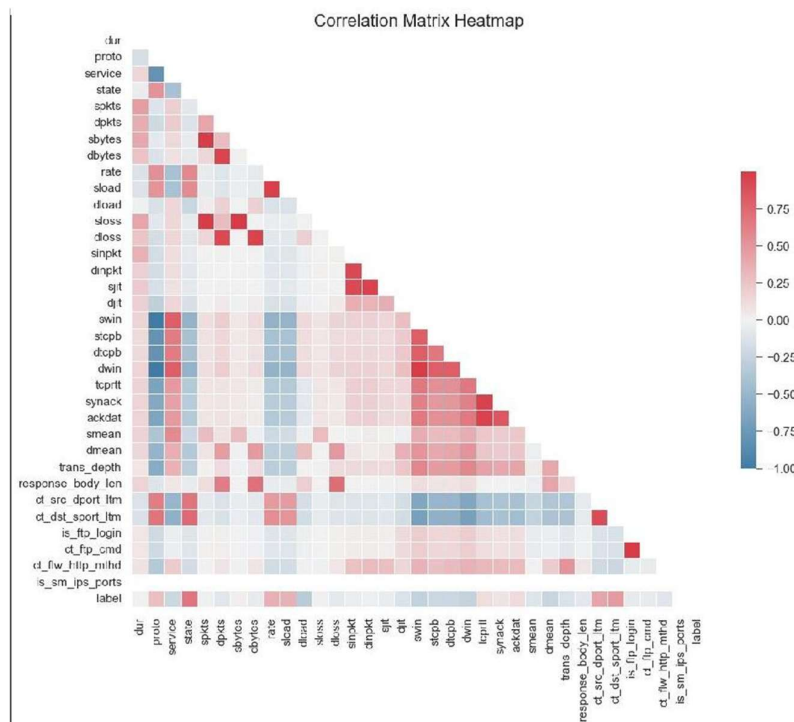


Figure 7: Correlation Matrix Heatmap

This figure shows the correlation between network traffic features using a color-coded heatmap.

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Results

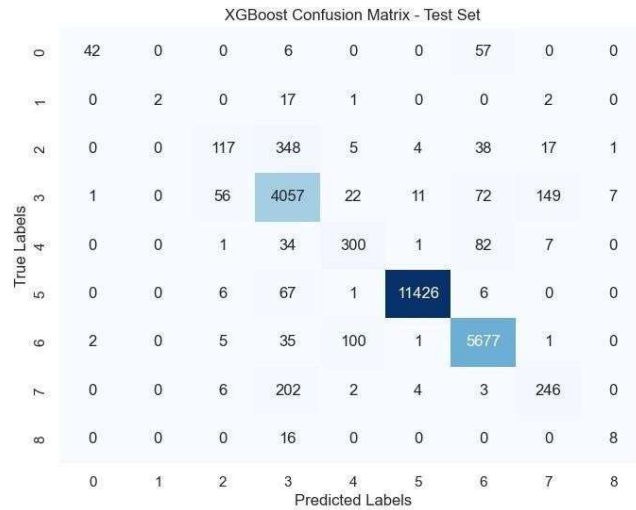


Figure 8: XGBoost Confusion Matrix

This figure displays the classification performance of the XGBoost model on the test set, showing true vs. predicted labels.

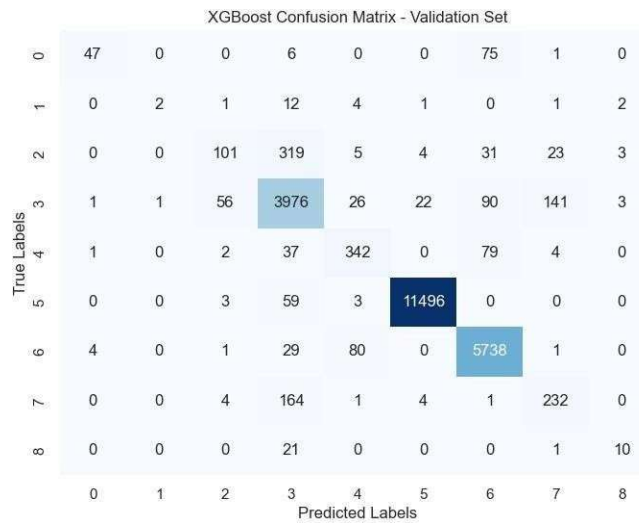


Figure 9: XGBoost Confusion Matrix (Validation Set)

This figure presents the validation performance of the XGBoost model, mapping predicted versus actual class labels.

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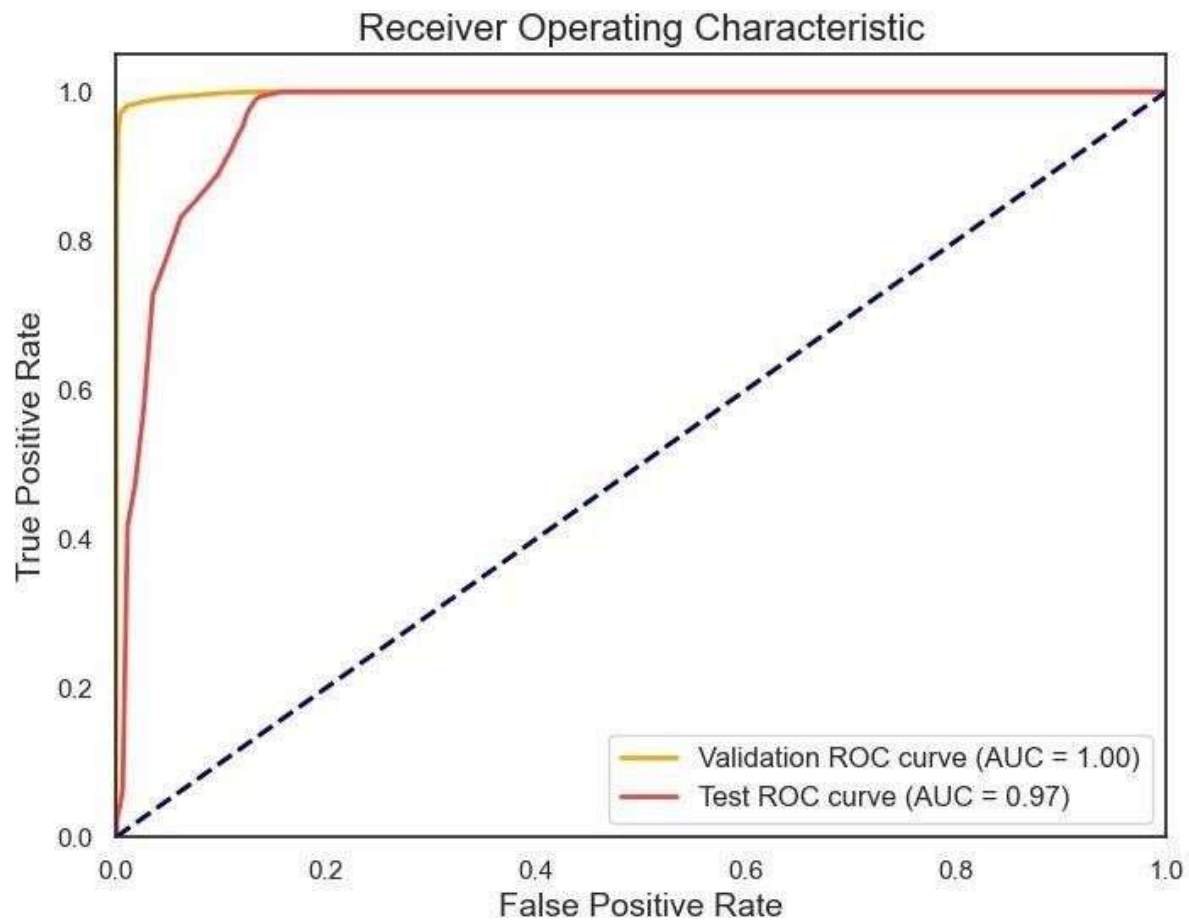


Figure 10: ROC Curve

This figure compares the ROC curves for validation and test sets, showing high model performance with AUC scores of 1.00 and 0.97, respectively.

Chapter 12

References

11.1 Conclusion

This project demonstrates the potential of Machine Learning, particularly the Random Forest algorithm, in addressing the cybersecurity challenges faced by IoT networks. By enabling precise and real-time detection of threats, the proposed system offers robust protection for IoT devices, securing data integrity and device functionality across various sectors. Through continuous adaptation to new attack patterns, this ML-based approach enhances the resilience of IoT infrastructure, making it a practical solution for the growing demand for IoT security. This work confirms that Machine Learning can provide a scalable and effective method for enhancing IoT security, thus supporting safer, more reliable connected systems.

11.2 Future Work

While the proposed system effectively demonstrates the potential of Machine Learning in securing IoT networks, future improvements can focus on several areas. Integrating **Federated Learning** would enhance privacy and scalability by training models across distributed IoT devices without transferring raw data to a central server. Incorporating **Explainable AI (XAI)** techniques would improve transparency, allowing security analysts to better understand why a model flagged certain behaviors as malicious, aiding trust and compliance. **Edge Computing Optimization** can reduce latency and ensure real-time threat detection by deploying lightweight models directly on IoT devices. Lastly, introducing **Adaptive Learning and Auto-Updates** would enable the model to automatically retrain on new threats and patterns, helping the system evolve to defend against emerging attacks.

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References

1. Baral, S., Saha, S., & Haque, A. (2024). An Adaptive End-to-End IoT Security Framework Using Explainable AI and LLMs. arXiv preprint. DOI: <https://doi.org/10.48550/arXiv.2409.13177>.
2. DeMedeiros, K., Hendawi, A., & Alvarez, M. (2023). A Survey of AI-Based Anomaly Detection in IoT and Sensor Networks. *Sensors*, 23(3), 1352. DOI: <https://doi.org/10.3390/s23031352>.MDPI
3. Ullah, F., Shah, M. A., & Khan, S. (2023). A Comprehensive Survey for IoT Security Datasets Taxonomy, Classification and Machine Learning Mechanisms. *Computers & Security*, 125, 102931. DOI: <https://doi.org/10.1016/j.cose.2023.102931>.ScienceDirect
4. Hussain, F., Abbas, S. G., & Hussain, S. (2023). Federated Deep Learning for Anomaly Detection in the Internet of Things. *Computers & Electrical Engineering*, 104, 108387. DOI: <https://doi.org/10.1016/j.compeleceng.2023.108387>.
5. Kumar, R., & Singh, R. (2023). Detection of Botnet in IoT Network Through Machine Learning Based Approach. *Journal of Ambient Intelligence and Humanized Computing*, 14, 1035–1050. DOI: <https://doi.org/10.1007/s41870-023-01603-1>
6. Ahmed, F., & Saeed, S. (2023). Enhancing IoT Network Security Through Deep Learning-Powered Intrusion Detection Systems. *Internet of Things and Cyber-Physical Systems*, 3, 100259. DOI: <https://doi.org/10.1016/j.iotcps.2023.100259>.
7. Li, Y., Chen, C., & Duan, M. (2023). Attention-Aware Encoder–Decoder Neural Networks for Heterogeneous Graphs of Things. *IEEE Transactions on Industrial Informatics*, 17(4), 2890–2898. DOI: <https://doi.org/10.1109/TII.2020.3009565>.MDPI
8. Aneja, S., Aneja, N., & Islam, M. S. (2018). IoT Device Fingerprint Using Deep Learning. In *Proceedings of the 2018 IEEE International Conference on Internet of Things and Intelligence System (IOTAIS)* (pp. 174–179). DOI: <https://doi.org/10.1109/IOTAIS.2018.8600848>.MDPI
9. Zhang, W., Zhang, Y., & Xu, L. (2019). Modeling IoT Equipment With Graph Neural Networks. *IEEE Access*, 7, 32754–32764. DOI: <https://doi.org/10.1109/ACCESS.2019.2903110>
10. Li, N., Hou, J., & Sha, L. (2005). Design and Analysis of an MST-Based Topology Control Algorithm. *IEEE Transactions on Wireless Communications*, 4(3), 1195–1206. DOI: <https://doi.org/10.1109/TWC.2005.846978>.MDPI
11. Phivou, P., Panousopoulou, A., & Tsakalides, P. (2015). On Realizing Distributed Topology Control in Low-Power IoT Platforms. In *Proceedings of the 2015 IEEE 2nd World Forum on Internet of Things (WF-IoT)* (pp. 98–103). DOI: <https://doi.org/10.1109/WF-IoT.2015.7389020>
12. Hussain, F., Hussain, R., Hassan, S. A. & Hossain, E. Machine learning in IoT security: Current solutions and future challenges. *IEEE Commun. Surv. Tutor.* 22(3), 1686–1721 (2020).
13. Shafiq, M., Tian, Z., Bashir, A. K., Du, X. & Guizani, M. Corrauc: A malicious BOT- IOT traffic detection method in IoT network using machine learning techniques. *IEEE Internet Things J.* 8(5), 3242–3254 (2020).

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References

13. Altunay, H. C. & Albayrak, Z. A hybrid CNN+ LSTM—Based intrusion detection system for industrial IoT networks. *Eng. Sci. Technol. Int. J.* 38, 101322 (2023).
14. Tahsien, S. M., Karimipour, H. & Spachos, P. Machine learning based solutions for security of Internet of Things (IoT): A survey. *J. Netw. Comput. Appl.* 161, 102630 (2020).
15. Abiodun, O. I., Abiodun, E. O., Alawida, M., Alkhawaldeh, R. S. & Arshad, H. A review on the security of the Internet of Things: Challenges and solutions. *Wirel. Person. Commun.* 119(3), 2603–2637 (2021).